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Image manipulation detection by multiple tampering traces and edge artifact enhancement

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ABSTRACT

Image manipulation detection has attracted considerable attention owing to the increasing security risks posed by fake images. Previous studies have proven that tampering traces hidden in images are essential for detecting manipulated regions. However, existing methods have limitations in generalization and the ability to tackle post-processing methods. This paper presents a novel Network to learn and Enhance Multiple tampering Traces (EMT-Net), including noise distribution and visual artifacts. For better generalization, EMT-Net extracts global and local noise features from noise maps using transformers and captures local visual artifacts from original RGB images using convolutional neural networks. Moreover, we enhance fused tampering traces using the proposed edge artifacts enhancement modules and edge supervision strategy to discover subtle edge artifacts hidden in images. Thus, EMT-Net can prevent the risks of losing slight visual clues against well-designed post-processing methods. Experimental results indicate that the proposed method can detect manipulated regions and outperform state-of-the-art approaches under comprehensive quantitative metrics and visual qualities. In addition, EMT-Net shows robustness when various post-processing methods further manipulate images.

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1. Introduction

Owing to the evolving image editing technologies and intelligent manipulation tools, more close-to-reality images can be easily produced. The misuse of digital image manipulation approaches, such as Internet scamming using fake faces [1], fake signature [2] and rumors [3], causes a trust crisis, considerable economic loss, and safety hazards to society.

Image manipulation techniques can be divided into two categories: homogenous and heterologous. Copy-move is the most common homogenous manipulation technique, whereas removal and splicing are widely used heterologous manipulation techniques. This study aims to detect manipulated regions in images forged by these three techniques (see Fig. 1), by which a large number of fake images have been produced. Locating manipulated regions of tampered images continues to be challenging. Sometimes, even careful observations are insufficient to identify finely manipulated areas. One reason is that types of image manipulation techniques are always unknown before detection, rendering developing a general detection method difficult. The other reason is that well-designed post-processing methods, such as local smoothing and image compression, can hide visual clues of manipulated regions.

Early approaches detect image manipulation by designing handcrafted features, such as color filter array (CFA) [4], noise inconsistency (NOI) [5], double quantization effect hidden among the discrete cosine transform (DCT) coefficients [6], local bidirectional coherency error [7] and error level analysis (ELA) [8]. Because of over-dependence on prior knowledge, accuracy, robustness, and generalization performance of these approaches are insufficient for detecting rapidly evolving manipulation techniques.

Recently, many Convolutional Neural Networks (CNN)-based methods have been proposed to detect manipulated regions at a pixel level. Most methods focus on detecting one specific technique, *i.e.,* heterologous manipulation [9,10] or homogenous manipulation [11,12].

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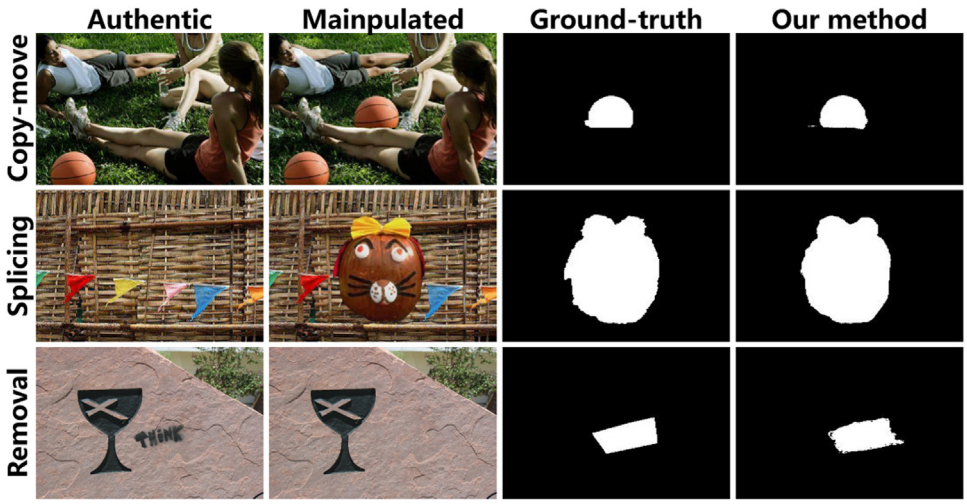

Fig. 1. Examples of manipulated images with different manipulation techniques, such as copy-move, splicing, and removal. The goal is to detect pixel-level binary masks of manipulated regions.

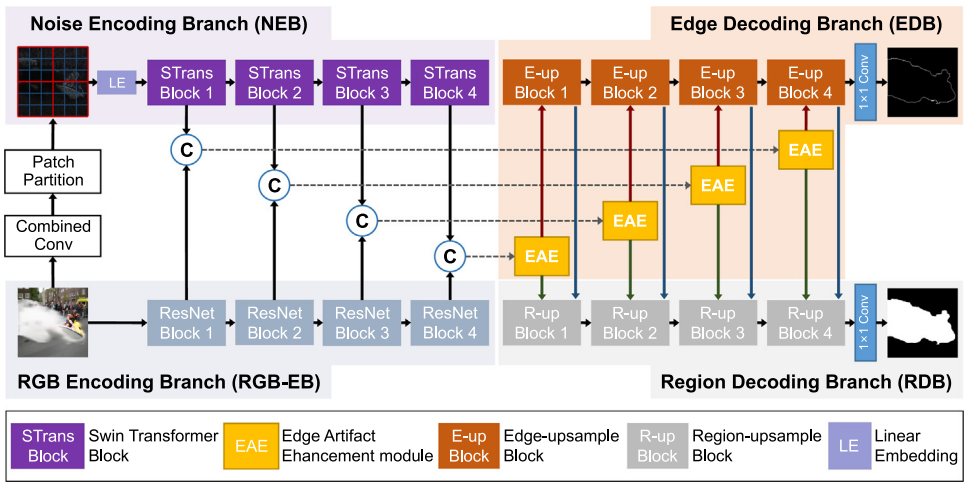

Fig. 2. Framework of our proposed EMT-Net for pixel-level manipulation detection. EMT-Net consists of a noise encoding branch, an RGB encoding branch, four edge artifact enhancement modules, an edge decoding branch, and a region decoding branch.

More recently, some CNN-based methods, such as ManTra [13], SPAN [14], or MVSS [15], offer generalized solutions. These methods extract tampering traces hidden in images to achieve superior performance and can be divided into two categories based on the type of tampering trace captured. The first category is based on noise maps [9,14] generated using special convolutional kernels to RGB images. These methods employ CNNs to extract abnormal local noise features from noise maps to distinguish heterologous regions but ignore the importance of global correlations. Unfortunately, homogenous manipulation techniques cause no local abnormality in noise maps, which makes it insufficient to be detected by local noise features. Therefore, local noise feature-based approaches have deficiencies in comprehensive detection. The second category captures edge artifacts of manipulated areas [15,16] using edge detection modules or edge supervision branches to improve edge feature extraction ability. However, it is challenging to distinguish boundary artifacts and edges of natural objects when visual artifacts are hidden by well-designed post-processing methods, such as local smoothing, image compression, and filtering.

We present a novel image manipulation detection Network by learning and Enhancing Multiple tampering Traces (EMT-Net), as shown in Fig. 2, to improve the generalization abilities and tackle post-processing approaches. EMT-Net focuses on detecting the three most common manipulating techniques, splicing, removal, and copy-move. First, a transformer-based Noise Encoding Branch (NEB) and CNN-based RGB Encoding Branch (RGB-EB) effectively extract and fuse multiple tampering traces, such as global noise, local noise, and artifact features. Second, proposed Edge Artifact Enhancement (EAE) modules and an edge supervision strategy in the Edge Decoding Branch (EDB) enhance subtle boundary artifacts of fused features for locating the edges of manipulated regions. Finally, a Region Decoding Branch (RDB) upsamples enhanced features for pixel-level prediction. Comprehensive experimental results on six benchmark datasets, *i.e.,* CASIA, NIST, Columbia, COVER, CoMoFoD, and DEFACTO, verify that our proposed method outperforms current state-of-the-art (SoTA) approaches even without pretraining.

Our main contributions can be summarized as follows:

- We simultaneously extract global and local noise features using transformers from noise maps in the proposed NEB to detect tampering traces produced by homologous and heterologous manipulation;
- We design EDB to reinforce the tampering traces at different scales. EDB combines proposed EAE modules and an edge supervision strategy to find edge artifacts of manipulated regions effectively even after applying post-processing methods;

- We present the EMT-Net, which can learn and enhance multiple tampering traces, including local noise inconsistency, global noise correlations, and subtle boundary artifacts from different scales. The fusion and enhancement of tampering traces enable precise detection of multiple content-changing manipulation techniques.

2. Related works

This section reviews the most relevant studies on CNN-based manipulation detection and localization methods. Then, we briefly introduce the transformer, one of the core components of the proposed network.

Current CNN-based approaches can be classified into those employing local noise and edge artifact features. The two categories are reviewed in the following subsections.

2.1. Local noise-based methods

Local noise features are extracted from noise maps modeled by residuals between original RGB images and RGB images estimated by interpolating methods [17]. Noise maps, which are always generated by applying special filters to original RGB images, can enhance tampering clues and suppress semantic information. RGB-N [17] employed steganalysis rich model (SRM) filters [18] to acquire noise maps and utilized CNNs to capture local noise abnormalities. Some approaches [15,19] used constrained Bayar convolutional blocks to learn image manipulation fingerprints and explore local inconsistencies from noise maps. ManTra [13], and SPAN [14] concatenated regular convolutional, SRM, and Bayar filters to extract local features from noise maps and RGB images. Li and Huang [9] designed a trainable pre-filtering module initialized with high-pass filters for enhancing tampered traces in detecting inpainted regions.

However, these noise map-based methods only used CNNs to extract local noise features without exploring global noise features. Local noise features cannot reveal manipulated regions forged by homologous manipulation techniques, weakening generalization performance. The proposed network extracts global and local noise features from noise maps using Swin transformers [20], thereby addressing heterologous and homogenous manipulation.

2.2. Edge artifact-based methods

Edge artifacts are vital clues for image manipulation detection as most manipulated region boundaries are surrounded by unnatural artifacts. Salloum et al. [16] proposed multi-task FCN (MFCN) to predict tampered areas and its boundary concurrently. Motivated by MFCN, Zhou et al. [21] designed a three-stage GSR-Net architecture including edge prediction, refinement, and segmentation branches. Supervised edge prediction and refinement branches improved detection results. More recently, Chen et al. [15] designed an edge-supervised branch using edge residual blocks in a shallow-to-deep manner to capture subtle boundary details. However, distinguishing the natural edges of objects and boundary artifacts becomes difficult when well-designed post-processing approaches (*e.g.,* local smoothing and noise adding) reduce visual clues. Therefore, we propose EDB, in which EAE modules and an edge supervision strategy can prevent the loss of subtle artifacts and learn more robust features for distinguishing edge artifacts and natural boundaries.

2.3. Transformer

The transformer [22] was first proposed for natural language processing tasks. Dosovitskiy et al. [23] introduced a vision transformer (ViT) to explicitly model global relationships between pixels and achieved impressive accuracy on several image processing tasks. However, ViT fails to capture subtle details of images owing to the limited local perception of patch-based self-attention modules. Liu et al. [20] proposed a more efficient and effective hierarchical Swin transformer using shifted windows-based self-attention. Unlike ViT, the Swin transformer can extract local and global features and achieve SoTA performance on various vision tasks, including object detection [24], and semantic segmentation [25]. Therefore, this study applies Swin transformers to capture global and local noise features from noise maps.

3. Method

This section briefly introduces the proposed method for detecting manipulation techniques and explains the motivation behind the study. Details of the four main network components are then discussed.

3.1. Overview and motivation

This work aims to predict pixel-level binary masks of manipulated regions in images using a novel model EMT-Net, which can extract and enhance sufficient tampering traces. We mainly detect three common forgery types, *i.e.,* copy-move, splicing, and removal. EMT-Net extracts and fuses multiple manipulation features, including global noise consistencies, local noise inconsistencies, and visual artifacts. The EMT-Net framework, illustrated in Fig. 2, consists of four main components: NEB, RGB-EB, EDB, and RDB. The EDB module enhances multiple features extracted using NEB and RGB-EB. Moreover, to prevent post-processing methods (*e.g.,* blurring, local smoothing, and compression) from decreasing visual clues, we propose an edge supervision strategy and EAE module to reinforce boundary artifacts of fused multiple features. After enhancement, RDB predicts manipulated regions. Moreover, edge details of features are gradually refined under the guidance of EDB during the RDB decoding process. Finally, RDB generates the binary masks of manipulated regions.

In the following sections, each main component is explicitly introduced.

3.2. Noise encoding branch

NEB aims to provide sufficient evidence for homologous and heterologous manipulation detection by extracting noise distribution features, including global noise consistencies and local noise inconsistencies. The NEB structure is shown in Fig. 2. We adopt a combined convolutional block [13,14] containing SRM filters [18], Bayar filters [19], and normal convolutional blocks to convert RGB images to noise maps. We then perform a patch partition layer to feed noise maps into the first Swin transformer block. As shown in Fig. 3(a), noise maps are partitioned into several windows of size $N_w \times N_w$ and each window is divided into patches of size $N_p \times N_p$. After partitioning, a linear embedding layer flattens the patches in each window. Finally, we apply Swin transformer blocks in series to extract global and local noise features from the flattened patches at different scales. The following subsections discuss details of the Swin transformer blocks.

3.2.1. Swin transformer block

As illustrated in Fig. 4, the Swin transformer block is made up of four layer-norm layers, two Multi-Layer Perceptrons (MLPs) [26], a window-based multi-head self-attention (W-MSA) module, a shifted window-based MSA (SW-MSA) module, and a patch merging layer. After each MLP and MSA module, a residual connection is established to prevent gradient vanishing. The l-th Swin

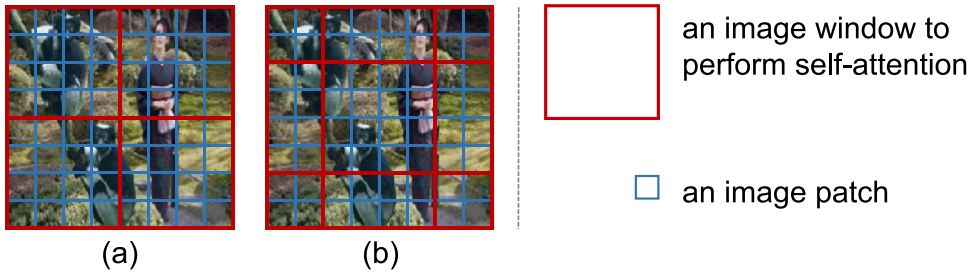

Fig. 3. Window partition. (a) Window and patch partition; (b) Shifted-window partition. All local windows in (a) are shifted two pixels in the right and bottom direction. After the shifting operation, nine new windows are generated.

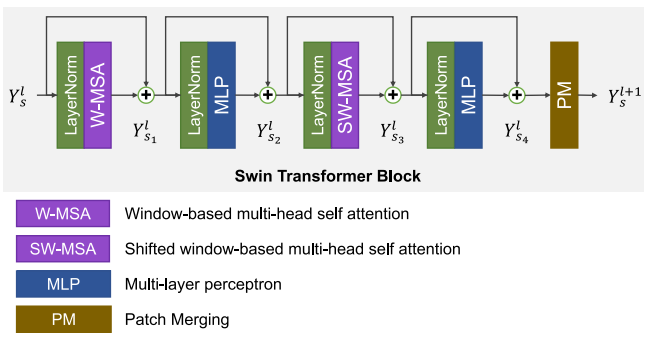

Fig. 4. Details of the Swin transformer block. (a) Framework of the Swin transformer block; (b) Architecture of the multi-layer perceptron in the Swin transformer block.

transformer block is formulated as follows:

$$Y^l_{s_1} = \mathrm{LN}(\text{W-MSA}(Y^l_s)) + Y^l_s, \tag{1}$$

$$Y^l_{s_2} = \mathrm{LN}(\mathrm{MLP}(Y^l_{s_1})) + Y^l_{s_1}, \tag{2}$$

$$Y^l_{s_3} = \mathrm{LN}(\text{SW-MSA}(Y^l_{s_2})) + Y^l_{s_2}, \tag{3}$$

$$Y^l_{s_4} = \mathrm{LN}(\mathrm{MLP}(Y^l_{s_3})) + Y^l_{s_3}, \tag{4}$$

$$Y^{l+1}_s = \mathrm{PM}(Y^l_{s_4}) \tag{5}$$

where $y^l_{s_1}$ and $Y^l_{s_4}$ are the input and output of the l-th Swin transformer block, respectively, $Y^l_{s_2}$ and $Y^l_{s_3}$ are the intermediate results, W-MSA$(\cdot)$ denotes W-MSA, SW-MSA$(\cdot)$ represents SW-MSA, MLP$(\cdot)$ denotes MLP, and PM$(\cdot)$ denotes the patch merging layer. These sub-components are discussed in the following paragraphs.

Multi-head self-attention. The W-MSA and SW-MSA modules are designed to extract global and local features from noise maps. Standard MSA module [23,27] only learns the global relationship of an image. To extract both local noise abnormalities and global noise consistencies, we use W-MSA to limit the global awareness across windows by limiting the MSA calculation process in each local image window (see Fig 3). Moreover, SW-MSA helps maintain the long-distance relationship learning capability by enlarging the receptive fields of W-MSA. W-MSA is computed as

$$\text{W-MSA}(Q, K, V) = V \cdot \mathrm{SoftMax}(B_{rp} + \frac{QK^T}{\sqrt{d}}) \tag{6}$$

where $B_{rp} \in \mathbb{R}^{N_p^2 \times N_p^2}$ is the relative position bias of each head of the self-attention module in computing process, $Q, K, V \in \mathbb{R}^{d \times N_p^2}$ are the *query, key* and *value* of patches, respectively, N_p^2 is the number of patches in the window to perform MSA, and d is the dimension of Q, and K. An example of window shifting is shown in Fig. 3(b).

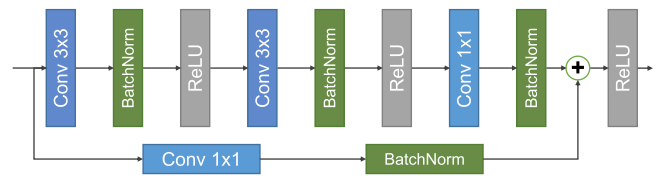

Fig. 5. Illustration of the ResNet block.

Patch merging. After performing the last MLP, the patch merging layer downsamples the feature maps. Thus, each group of 2×2 adjacent patches is separated into four new feature maps. The resolution of the output feature maps is downsampled by 2 times, and the number of channels is increased by 4 times. Finally, we resize the Swin transformer block outputs $x^l_{s_4}$ to match that of ResNet blocks in RGB-EB at corresponding scales.

3.3. RGB encoding branch

RGB-EB aims to find unnatural visual artifacts existing in original RGB images. As shown in Fig. 2, four ResNet RGB-EB blocks are connected in sequence (based on ResNet50 [28]). The ResNet block structure is shown in Fig. 5. A CNN-based backbone is used to derive detailed local artifact features from RGB images. Each ResNet block reduces the resolution by 2 times. Furthermore, we adopt a concatenation operation to fuse the same resolution features extracted from NEB and RGB-EB.

3.4. Edge decoding branch

To better capture the edge details and assist model predictions, we design an EDB for edge artifacts learning. As shown in Fig. 2, EDB consists of four EAE modules, four Edge-upsample (E-up) blocks, and an edge supervision strategy. Supervised by edges of manipulated regions, E-up blocks recover the resolution using enhanced tampering feature maps from EAE modules. After channel compression by 1×1 convolutional layer, outputs of the last E-up block represent binary boundary masks of manipulated regions. Four E-up blocks are cascaded to reach the full resolution from $\frac{H}{16} \times \frac{W}{16}$ to $H \times W$. Proposed EAE modules, E-up blocks, and edge supervision strategy are detailed in the following subsections.

3.4.1. Edge artifact enhancement module

Post-processing methods can weaken the edge artifacts hidden in images, making it difficult to find subtle artifacts directly. Consequently, as shown in Fig. 6(a), we propose CNN-based EAE modules to indirectly enhance artifacts of fused feature maps. In each EAE module, the convolution layers are forced to learn to eliminate

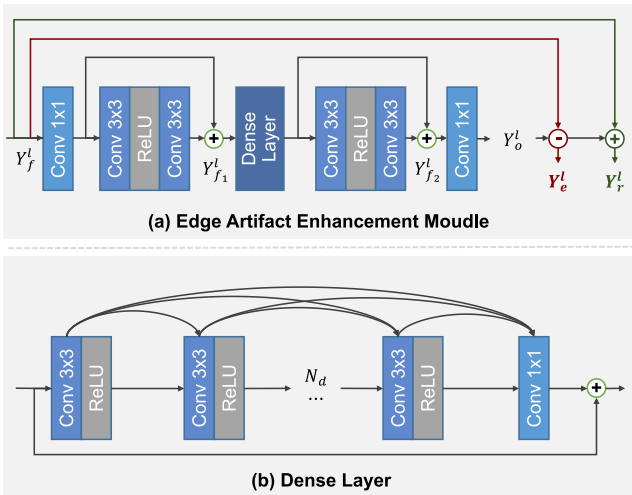

Fig. 6. Framework of (a) EAE module; (b) dense layer in the EAE module.

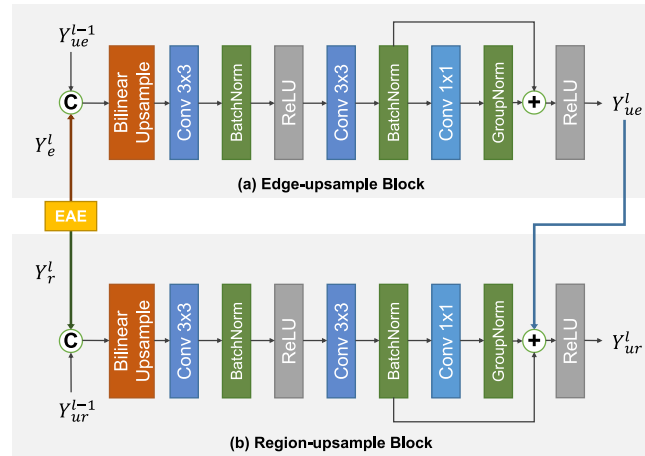

Fig. 7. Overall structure and relationship of edge-upsample (E-up) and region-upsample (R-up) blocks. (a) E-up block framework; (b) R-up block framework.

edge artifacts surrounding manipulated regions. Thus, the residuals between the inputs of the EAE module and eliminated feature maps can represent edge artifact features. The proposed EAE module consists of two 1×1 convolutional layers, two residual groups (RG) layers, and a dense layer (DL). The 1×1 convolutional layers vary the dimension of the input fused tampering features. The residual group layer has two 1×1 convolutional layers and a ReLU activation layer. Each EAE module has one input Y_f^l and two outputs Y_e^l, and Y_r^l. The Y_f^l represents fused feature maps from NEB and RGB-EB at the l-th scale. Y_e^l denotes the features of boundary artifacts, named edge output, and Y_r^l denotes the feature maps with enhanced edge artifacts, called region output. The EAE module is then formulated as follows:

$$Y_{f_1}^l = \text{Conv } 1\text{x}1(\text{RG}(Y_f^l)), \tag{7}$$

$$Y_{f_2}^l = \text{DL}(\text{RG}(Y_{f_1}^l)), \tag{8}$$

$$Y_o^l = \text{Conv } 1\text{x}1(Y_{f_2}^l), \tag{9}$$

$$Y_e^l = Y_f^l - Y_o^l, \tag{10}$$

$$Y_r^l = Y_f^l + Y_e^l \tag{11}$$

where RG is the residual group layer (computed using Eq. (12)), and DL denotes the dense layer shown in Fig. 6(b).

$$\text{RG}(Y) = \text{Conv } 3 \times 3(\text{ReLU}(\text{Conv } 3 \times 3(Y))) + Y \tag{12}$$

3.4.2. Edge-upsample block

The framework of the E-up block is shown in Fig. 7(a). In each E-up block in EDB, hidden feature maps Y_{ue}^{l-1} from the previous block are concatenated with edge outputs Y_e^l of the corresponding EAE module (the hidden feature maps are not available in E-up block 1). Next, a bilinear upsample doubles the resolution of concatenated features. Next, two successive 3×3 convolutional layer groups joined with a residual 1×1 convolutional layer group recover subtle edge artifacts and aggregate feature maps.

3.4.3. Edge supervision strategy

An edge supervision strategy that forces the EDB to concentrate on subtle edge artifacts can effectively distinguish the natural edges of objects and edge artifacts. We use boundaries around the manipulated regions to train the whole EDB. Dice loss is used for edge supervision because the proportion of boundary pixels around the manipulated regions is small. It is widely used for training deep learning-based networks with highly imbalanced data. This loss ignores the unimportant background information and uses a set similarity of the foreground to perform evaluation, which is suitable for edge supervision tasks. The loss function of the edge supervision is

$$\text{loss}_e(x) = 1 - \frac{2 \cdot \sum_{i=1}^{H \times W} E(x_i) \cdot y_i}{\sum_{i=1}^{H \times W} E(x_i)^2 + \sum_{i=1}^{H \times W} y_i^2}, \tag{13}$$

where $E(x_i)$ is the prediction of EDB and implies the probability that the i-th pixel is manipulated boundaries in image x, and y_i denotes the binary edge ground-truth indicating whether the i-th pixel belongs to boundaries of manipulated regions.

3.5. Region decoding branch

RDB predicts the manipulated regions by upsampling the fused tampering traces, the boundary artifacts of which are strengthened by EAE modules and E-up blocks in the EDB. There is no considerable difference between a Region-upsample (R-up) block and an E-up block (Fig. 7(b)). R-up blocks in the RDB receive region outputs Y_r^l from EAE modules. Moreover, edge artifact features are further enhanced by adding each E-up block output Y_{ue}^l in the EDB at the same scale. The enhanced edge features guide the precise detection of the boundary and region details of manipulated areas.

We apply binary cross entropy as a loss function in this branch. The region loss function considering manipulated and authentic pixel losses is as follows:

$$\text{loss}_r(x) = -\frac{\sum_{i=1}^{H \times W} [y_i \ln R(x_i) + (1 - y_i)\ln(1 - R(x_i))]}{H \times W}, \tag{14}$$

The total loss of the proposed EMT-Net consists of the edge and region losses and is formulated as

$$\text{loss}_t(x) = \gamma_e \cdot \text{loss}_e(x) + \gamma_r \cdot \text{loss}_r(x), \tag{15}$$

where γ_e and γ_r are the weights of the edge and region losses, respectively. Function $R(\cdot)$ denotes pixel-level results of manipulated regions predicted by RDB.

Table 1
Description of the datasets .

Dataset name	Training-testing split		Involved manipulation technique			Post-processed
	Training	Testing	Splicing	Copy-move	Removal	
CASIA [29]	5123	921	✓	×	✓	✓
NIST [30]	404	160	✓	✓	✓	✓
Columbia [31]	\	180	✓	×	×	×
COVER [32]	75	25	×	✓	×	✓
CoMoFoD [33]	3500	1500	×	✓	×	✓
DEFACTO [34]	103,173	44,217	✓	✓	✓	✓

4. Experimental results

This section first introduces the experimental setup. A comparison of the proposed method with six SoTA approaches on six datasets is then presented. In addition, an ablation study verifies the effectiveness of the main components in EMT-Net. Finally, we evaluate the robustness of EMT-Net when dealing with various post-processing approaches.

4.1. Experiment setup

We describe the setup of the following experiments in this section, including introductions of datasets, evaluation metrics, and implementation details.

4.1.1. Datasets

The six image manipulation detection datasets used are CASIA [29], NIST [30], Columbia [31] (use CASIA for training), COVER [32], CoMoFoD [33], and DEFACTO [34]. Details of these datasets are presented in Table 1. For a fair comparison, we adopt the most popular training-testing splitting configurations on CASIA, NIST, COVER, and Columbia, as in [14,17,19]. For CoMoFoD and DEFACTO, we choose a 7:3 training-testing ratio.

4.1.2. Evaluation metrics

Following previous works [14,15,17,19], two commonly used pixel-wise classification metrics were adopted to evaluate the performance: F1-score (F1) and Area Under the receiver operating characteristic Curve (AUC). In this work, AUC is formulated as

$$\mathrm{AUC} = \frac{\sum_{i\in A, j\in M} I(R(x_i), R(x_j))}{|A| \times |M|} \tag{16}$$

$$I(a,b) = \begin{cases} 1, & a < b \\ 0.5, & a = b \\ 0, & a > b \end{cases} \tag{17}$$

where A and M are the set of all authentic and manipulated pixels, respectively, $R(x_i)$ and $R(x_j)$ denote the prediction values of the i-th and j-th pixels, respectively. F1 is calculated as

$$\mathrm{F1} = \frac{2 \cdot \mathrm{TP}}{2 \cdot \mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{18}$$

where TP are the number of correctly detected manipulated pixels, FP and FN are the number of manipulated and authentic pixels mistakenly predicted, respectively. Here, a method with high AUC will not always misjudge pixels as manipulated pixels, whereas high F1 denotes that the method detects the most manipulated pixels. F1 and AUC are calculated on each image independently. The mean value of each metric is obtained to make comparisons in the following experiments.

4.1.3. Implementation details

Our approach is implemented based on PyTorch. The model is trained on a single RTX 3090 GPU with a batch size of 16. Because of the memory limitations of the computing device, we resize all input images to 256×256, following [13]. An Adam optimizer is used with a learning rate of 4×10^{-5} and a decay of 1×10^{-3}. In our loss function, the edge loss weight $\gamma_e = 0.3$ and region loss weight $\gamma_r = 0.7$. The number of 3×3 convolutional layer in the dense layer is $N_d = 2$. The window and patch sizes of Swin transformer blocks are 8 and 2, respectively.

4.2. Comparison with sotas

In this section, we first make a brief introduction of compared methods. Then, we show quantitative and qualitative results of experiments on six datasets.

4.2.1. Compared methods

We evaluate and compare EMT-Net with six SoTA methods *i.e.,* ManTra [13], SPAN [14], MVSS [15], GSRNet [21], DenseFCN [35], and LocateNet [36]. These compared methods have publicly available source codes. For a fair and reproducible comparison, all parameters involved in the compared methods are optimally assigned or automatically chosen as described in the reference papers.

4.2.2. Quantitative results

We evaluate and compare EMT-Net with six CNN-based SoTA methods on six datasets. The comparison of AUC and F1 is reported in Table 2. The proposed EMT-Net has the best overall performance even without pre-training and fine-tuning (given in bold for each dataset), revealing the superiority of fusing and enhancing multiple tampering features. More specifically, global noise features, ignored in SoTA methods, guarantee the generalization ability of EMT-Net. Moreover, the enhancement of subtle edge artifacts ensures the fineness of predicted binary masks, whereas other methods fail to employ edge information or only develop low-robust edge artifact extraction modules.

4.2.3. Qualitative results

CNN-based manipulation detection results are visualized in Fig. 8. The detection results when using EMT-Net are consistently accurate regardless of the type of image manipulation technique. Better qualitative results of EMT-Net can be attributed to extensive exploration of tampering features and explicit reinforcement of subtle artifacts. In column 4 of Fig. 8, the SoTA methods generate many false alarms or incomplete predictions in images manipulated by homologous techniques because the global noise features are ignored when locating highly similar areas. The boundary detection results of ManTra, SPAN, DenseFCN, and LocateNet are far less precise than that of EMT-Net, because they neglect edge artifact features. In contrast, MVSS and GSRNet employ edge supervision branches to learn edge information, however, EMT-Net offers improved precision results when dealing with post-processing approaches (especially the second rows in Fig. 8). The edge predictions using EMT-Net are close to manipulated region boundaries,

Table 2
Quantitative comparison of EMT-Net with the six SoTA methods on CASIA, NIST, Columbia, COVER, CoMoFoD, and DEFACTO.

Method	CASIA		NIST		Columbia		COVER		CoMoFoD		DEFACTO	
	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
ManTra	0.796	0.267	0.959	0.638	0.736	0.243	0.777	0.283	0.900	0.545	0.638	0.045
SPAN	0.709	0.213	0.779	0.252	0.741	0.463	0.791	0.325	0.854	0.267	0.869	0.217
MVSS	0.847	0.318	0.981	0.768	0.808	0.417	0.808	0.284	0.889	0.476	0.932	0.418
GSRNet	0.836	0.340	0.967	0.640	**0.900**	0.433	0.788	0.218	0.867	0.492	0.880	0.250
DenseFCN	0.809	0.203	0.979	0.812	0.761	0.257	0.754	0.185	0.889	0.331	0.910	0.404
LocateNet	0.754	0.273	0.986	0.738	0.718	0.411	**0.813**	0.282	0.897	0.590	0.941	0.457
EMT-Net	**0.856**	**0.459**	**0.987**	**0.825**	0.832	**0.561**	0.812	**0.353**	**0.906**	**0.594**	**0.942**	**0.481**

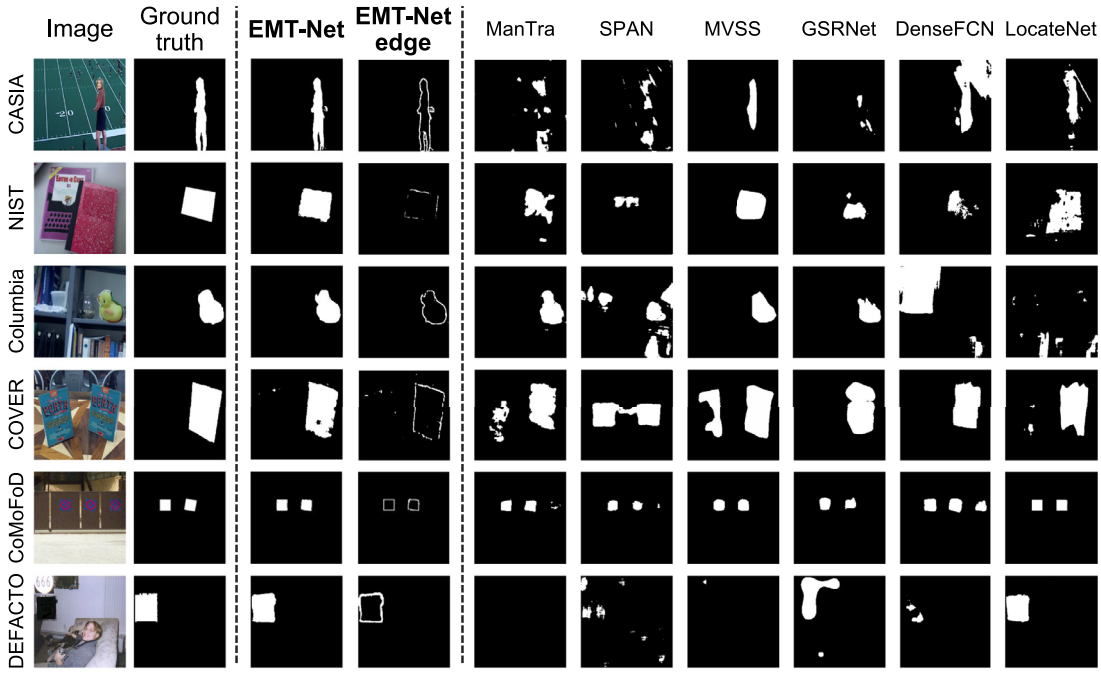

Fig. 8. Sample qualitative results of the proposed EMT-Net compared with six SoTA methods on six datasets. From left to right, we show manipulated images, ground-truth binary masks of manipulation, predictions of the proposed EMT-Net, the EDB of our EMT-Net, MVSS, ManTra, SPAN, GSRNet, DenseFCN, and LocateNet.

indicating that the proposed supervision and enhancement strategies successfully extract subtle artifacts hidden in manipulated images. Edge artifact features guide EMT-Net in locating manipulated regions for improved pixel-level detection.

4.3. Ablation study

We conduct an ablation study on NEB, RGB-EB, and EDB components to investigate their contribution to improving the EMT-Net's performance using the same metrics as in the previous experiment, *i.e.*, AUC, and F1. For a comprehensive evaluation, we design 10 setups, and the core branches of each setup are shown in Table 3. The results of all setups are provided on the most challenging dataset, CASIA [29].

4.3.1. Effect of NEB

Three setups are considered to evaluate the contribution of NEB. N-A only uses ResNet blocks to extract local noise features, N-B merely adopts ViTs to explore global noise features, and N-C makes predictions without using any noise features from NEB. As shown in Table 3 and Fig. 9(a), the effectiveness of local noise features can be proved by quantitative and qualitative comparisons between setups N-A and N-C. Specifically, AUC increases from 0.791 to 0.815, and F1 increase from 0.304 to 0.325. The performance improvements demonstrate that noise anomalies can help locate tampered areas. In addition, when comparing N-B and setup N-C, the global noise feature effectively improves the model performance due to patch-level pixel self-attention. Unlike N-B and N-C, EMT-Net extracts global and local noise features using Swin transformer blocks, taking advantage of the complementarity between tampering traces. Moreover, the visualization results in Fig. 9(a) verify that the predictions of full EMT-Net are more accurate, again proving the importance of extracting global and local features in NEB.

4.3.2. Effect of RGB-EB

Although the structure of ResNet blocks in RGB-EB is simple, Table 3 and Fig. 9(b) show that it considerably improves the overall results. Comparing our full EMT-Net with setups R-D (employing Swin transformer blocks in RGB-EB), R-E (adopting ViTs in RGB-EB), and R-F (removing the whole RGB-EB branch) show reduced AUC and F1. The visualization results of R-D, R-E, and R-F are far from ground truths. Hence, depending on noise features extracted by NEB is insufficient for precise detection and extracting local visual artifacts using CNN from RGB images is important. ResNet blocks in RGB-EB can extract local information of manipulated regions more finely. Swin transformers and ViTs fail to extract subtle local features, which leads to the insufficient ability in locating manipulated regions. The results indicate combining the features from RGB-EB and NEB can maximize the detection performance.

Table 3
Performance comparison of EMT-Net setups used in the ablation study. Each setup is trained on CASIA v2 and tested on CASIA v1. A benchmark training strategy is adopted for all setups. *global + local* denotes global and local features extracted by Swin transformer blocks, *global* represents global features extracted by ViTs, *local* denotes local feature captured by ResNet blocks, *ES* is the edge supervision, *EAE* is the edge artifact enhancement module, and *ESB* is the edge supervise branch proposed in [15].

Ablation subject	Setup name	Core branch			AUC	F1
		NEB	RGB-EB	EDB		
NEB	N-A	local	local	ES + EAE	0.815	0.325
	N-B	global	local	ES + EAE	0.826	0.362
	N-C	×	local	ES + EAE	0.791	0.304
RGB-EB	R-D	global + local	global + local	ES + EAE	0.651	0.133
	R-E	global + local	global	ES + EAE	0.681	0.083
	R-F	global + local	×	ES + EAE	0.650	0.063
EDB	E-G	global + local	local	ESB [15]	0.807	0.366
	E-H	global + local	local	×	0.802	0.345
	E-I	global + local	local	ES	0.835	0.375
-	**full**	**global + local**	**local**	**ES + EAE**	**0.856**	**0.459**

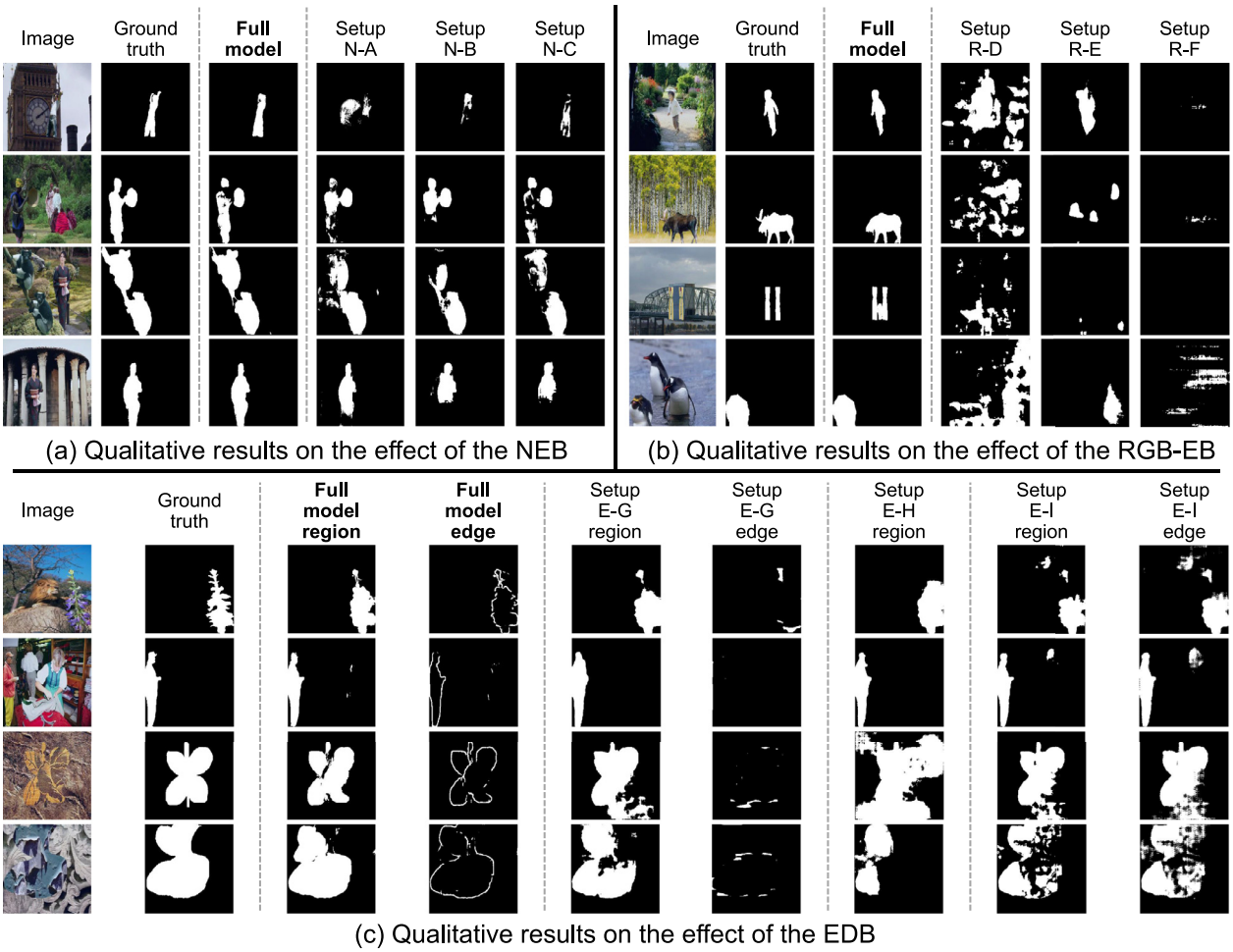

Fig. 9. Examples of qualitative manipulation detection results in the ablation study on the effectiveness of two encoding branches. The comparison between different settings of (a) NEB, (b) RGB-EB, and (c) EDB.

4.3.3. Effect of EDB

Setup E-G utilizes the Edge Supervise Branch (ESB) from MVSS [15] instead of our proposed EDB to extract edge features by applying handcrafted feature-based CNN blocks directly. E-H eliminates EDB, whereas E-I retains the edge supervision strategy without EAE modules. However, as shown in Table 3 and Fig. 9(c), E-G, E-H, and E-I contribute to lower performance than the EMT-Net. EMT-Net can predict almost all details of edges around manipulated regions, whereas E-G only detects part of boundaries (compare columns 4 and 6 in Fig. 9(c)). Hence, when detecting post-processed manipulated images, EAE modules can better distinguish natural boundaries and subtle edge artifacts using residual features compared to directly extracting edge features like the ESB. Without the assistance of EAE modules (column 9 in Fig. 8), edge supervision can hardly detect edge artifacts (column 9 in Fig. 9(c)). Combining the edge supervision and EAE modules, EMT-Net can precisely find visual artifacts around manipulated regions, even with post-processed images. EDB gradually uses the enhanced edge artifacts

Table 4
Robustness comparison of the proposed method with SoTA methods over NIST dataset. The evaluation metric is AUC.

Post-processing method	Parameter value	Methods						
		Ours	MVSS	SPAN	ManTra	GSRNet	DenseFCN	LocateNet
-	-	**0.987**	0.981	0.779	0.959	0.967	0.979	0.985
Gaussian blur	kernel=3	**0.987**	0.982	0.777	0.732	0.965	0.979	0.984
	kernel=9	**0.985**	0.980	0.771	0.639	0.958	0.976	0.983
	kernel=15	**0.982**	0.978	0.765	0.643	0.948	0.970	0.981
JPEG compression	quality=100	**0.987**	0.981	0.778	0.911	0.948	0.979	0.985
	quality=75	**0.987**	0.981	0.778	0.761	0.948	0.979	0.985
	quality=50	**0.986**	0.980	0.777	0.725	0.944	0.979	0.984
Gaussian noise	sigma=3	**0.986**	0.981	0.708	0.600	0.965	0.979	0.982
	sigma=9	**0.983**	0.980	0.707	0.605	0.962	0.976	0.979
	sigma=15	**0.983**	0.980	0.705	0.597	0.961	0.970	0.975

Table 5
Robustness comparison of the proposed method with SoTA methods over NIST dataset. The evaluation metric is F1.

Post-processing method	Parameter value	Methods						
		Ours	MVSS	SPAN	ManTra	GSRNet	DenseFCN	LocateNet
-	-	**0.825**	0.768	0.252	0.638	0.640	0.812	0.738
Gaussian blur	kernel=3	**0.824**	0.768	0.249	0.198	0.602	0.802	0.737
	kernel=9	**0.811**	0.742	0.239	0.165	0.583	0.749	0.716
	kernel=15	**0.782**	0.711	0.231	0.161	0.573	0.690	0.645
JPEG compression	quality=100	**0.825**	0.766	0.252	0.457	0.572	0.811	0.738
	quality=75	**0.825**	0.770	0.252	0.226	0.569	0.812	0.736
	quality=50	**0.812**	0.761	0.250	0.186	0.567	0.811	0.691
Gaussian noise	sigma=3	**0.822**	0.764	0.164	0.060	0.602	0.802	0.724
	sigma=9	**0.809**	0.758	0.165	0.051	0.603	0.749	0.707
	sigma=15	**0.808**	0.751	0.165	0.050	0.597	0.690	0.682

features to guide the better detection of the tampered area during the upsampling process. The results validate that the proposed EDB with EAE modules can improve performance remarkably.

4.4. Robustness analysis

We separately apply post-processing methods, Gaussian blur, JPEG compression, and Gaussian noise on NIST [30] to verify the robustness of EMT-Net. For each post-processing method, we vary the kernel size in Gaussian blur (from 3 to 15), quality in JPEG compression (from 50% to 100%), and variance of Gaussian noise (from 3 to 15) for comprehensive verification. Results of robustness analysis are shown in Tables 4 and 5. Compared with SoTA CNN-based methods, EMT-Net achieves the most general robust performance using multiple enhanced tampering traces. The performance of SoTA methods drops when dealing with post-processed images, because of the failure to extract weakened edge artifacts. Moreover, the combination of EDB and EAE modules can recover boundary details hidden by post-processing methods. In summary, our strategy of fusing different tampering features (by NEB and RGB-EB) and reinforcing subtle artifacts (by edge supervision and EAE modules) can effectively tackle the challenges bought by post-processing methods.

4.5. Limitation analysis

We discuss the limitations of our EMT-Net in two special failure cases shown in Fig. 10.

In the first case, in Fig. 10(a), EMT-Net makes precise predictions on images forged by advanced Generative Adversarial Networks (GAN), *i.e.*, GAN inversion. GAN inversion helps generate close-to-reality images by inverting images back to the latent space

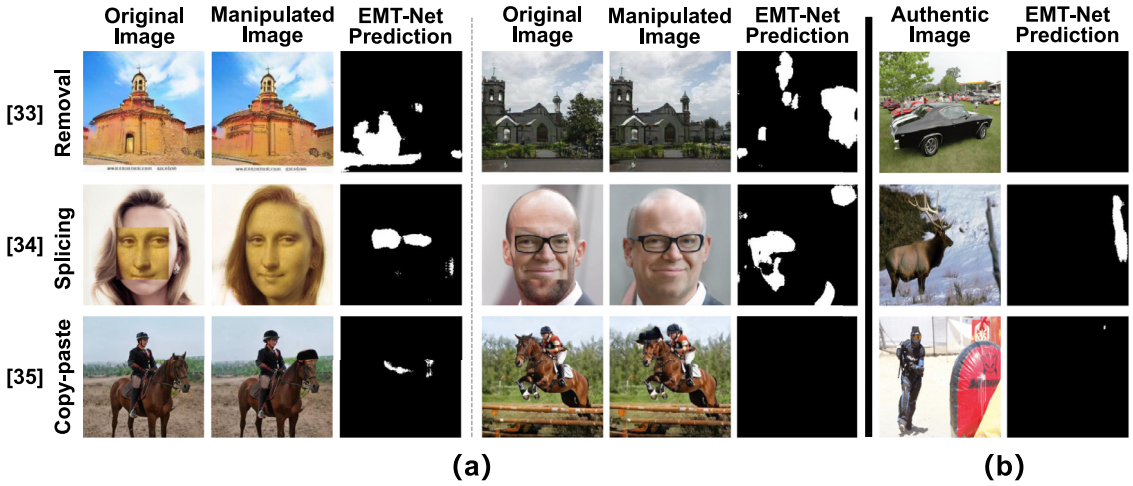

Fig. 10. Two failure cases of the proposed EMT-Net. (a) Images forged by GAN inversion-based techniques. (b) Authentic images without manipulated pixels.

of pretrained GANs and reconstructing them based on generators with inverted codes [37]. Bau et al. [38] removed artifact-causing modules in GANs by quantifying the causal relationship between modules. Recent GAN inversion-based methods can produce high-quality manipulated images. Zhu et al. [39] developed an in-domain GAN inversion approach and a domain-regularized optimization method to better support close-to-reality editing by varying the inverted codes. Bau et al. [40] proposed a method to rewrite pretrained GANs and generate realistic images by modifying the learned rules of GANs. We examine the proposed EMT-Net on images forged by GAN inversion-based removal [38], splicing [39], and copy-paste [40]. When the artifacts in tampered regions are strongly reduced, the proposed method finds incomplete tampered areas owing to insufficient tampering traces. Moreover, GAN-based manipulated techniques can harmonize heterogeneous regions, making detection difficult.

The second case is the performance drop of EMT-Net when dealing with authentic images without manipulated regions. In Fig. 10(b), though the proposed method can distinguish authentic images (first row), false alarms are present in the second and third rows when semantic objects are locally inconsistent due to out-of-focus.

In summary, the proposed method shows good performance only when locating non-GAN-based manipulated areas in tampered images.

5. Conclusion

This paper presents a novel EMT-Net that learns multiple tampering traces from noise maps and RGB images for image manipulation detection. EMT-Net fuses tampering features, *i.e.,* both global and local noise features learned by Swin transformers, and detail artifact features captured by CNN to detect the image manipulation techniques. The EDB can strengthen subtle boundary artifacts of fused feature maps and prevent the loss of artifacts weakened by post-processing methods. Extensive experiments demonstrate that EMT-Net outperforms SoTA approaches. Although our approach achieves superiority and robustness against various post-processing methods, it has some limitations, including low efficiency and high video memory usage. We also notice the drawbacks of the EMT-Net in dealing with advanced GAN and authentic images, which may be solved by more discriminative prior knowledge and appropriate loss functions, respectively. Furthermore, our method cannot accurately detect images forged by unlearned manipulation or post-processing techniques. A global feature extractor more efficient than Swin transformers is desirable in the future. Moreover, exploring uncertainty estimation methods to help detect out-of-distribution manipulated images will be an exciting new direction.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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